

Problem 10.1 A mild steel tube 5 meters long, 30 mm internal diameter and 4 mm thick is used as a strut with both ends hinged. Find the collapsing load. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Moment of inertia of section

$$I = \frac{\pi}{64} (D^4 - d^4) = \frac{\pi}{64} [38^4 - 30^4] = 62593 \text{ mm}^4$$

Since both ends of the column are hinged

$$\therefore \text{Effective length} \quad L = 5 \text{ m} = 5000 \text{ mm}$$

$$\therefore \text{Crippling load} \quad P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times 62593}{(5000)^2} = 5189.12 \text{ N}$$

Problem 10.2 A strut 2.50 meters long is 60 mm in diameter. One end of the strut is fixed while its other end is hinged. Find the safe compressive load for the member using Euler's formula, allowing a factor of safety of 4. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution: Since one end of the strut is fixed and the other end is hinged, effective length of strut

$$= L = \frac{l}{\sqrt{2}} = \frac{2500}{\sqrt{2}} = 1768 \text{ mm}$$

$$\text{Moment of inertia of the section} \quad I = \frac{\pi d^4}{64} = \frac{\pi \times 60^4}{64} = 636172 \text{ mm}^4$$

$$\therefore \text{Crippling load} \quad = P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times 636172}{1768^2} = 421820 \text{ N}$$

$$\therefore \text{Safe load} \quad = \frac{\text{Crippling load}}{\text{Factor of safety}} = \frac{421820}{4} = 105455 \text{ N}$$

Problem 10.3 A solid circular compression member 50 mm in diameter is to be replaced by a hollow circular section of the same material. Find the size of the hollow section if the internal diameter is 0.6 times the external diameter.

Solution: Diameter of the solid section $d = 50 \text{ mm}$

External diameter of the hollow section d_1 ,

Internal diameter of the hollow section $d_2 = 0.6 d_1$

For the two columns to have the same strength

$$I_{\text{hollow}} = I_{\text{solid}}$$

$$\frac{\pi}{64} [d_1^4 - 0.6^4 d_1^4] = \frac{\pi}{64} d^4$$

$$d_1^4 - 0.6^4 d_1^4 = d^4$$

$$d_1^4 = \frac{d^4}{1 - 0.6^4}$$

$$d_1^4 = \frac{d^4}{0.8704} = \frac{50^4}{0.8704}$$

∴

$$d_1 = 51.765 \text{ mm}$$

$$d_2 = 0.6 \times 51.765 = 31.059 \text{ mm}$$

Problem 10.4 Find the shortest length L for pin ended steel column having a cross-section of 60 mm × 100 mm for which Euler's formula applies. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$ and critical proportional limit is 200 N/mm^2 .

Solution: Least moment of inertia of the section

$$= \frac{100 \times 60^3}{12} = 18 \times 10^5 \text{ mm}^4$$

$$\text{Euler's critical load} = P = \frac{\pi^2 EI}{L^2}$$

Stress corresponding to Euler's critical load

$$= \frac{\pi^2 EI}{AL^2}$$

When this stress reaches 200 N/mm^2

$$\frac{\pi^2 EI}{AL^2} = 200$$

$$L^2 = \frac{\pi^2 EI}{A \times 200} = \frac{\pi^2 \times 2 \times 10^5 \times 18 \times 10^5}{(60 \times 100) \times 200} = 2.961 \times 10^6$$

$$L = 1720.72 \text{ mm}$$

Problem 10.5 Calculate the critical load for a strut which is made of a bar circular in section and 6 meter long and which is pin-jointed at both ends. The same bar when freely supported gives a mid-span deflection of 10 mm under a load of 80 N at the centre.

Solution: Analysis as a beam

$$l = 6000 \text{ mm}, \quad W = 80 \text{ N}, \quad \delta = 10 \text{ mm},$$

$$\delta = \frac{Wl^3}{48 EI}$$

∴

$$EI = \frac{Wl^3}{48\delta} = \frac{80 \times 6000^3}{48 \times 10} = 36 \times 10^9 \text{ N/mm}^2$$

Analysis as a strut

$$\text{Euler's critical load} \quad P = \frac{\pi^2 EI}{l^2} = \pi^2 \times 36 \times 10^9 \times \frac{1}{6000^2} = 9859.6 \text{ N}$$

Problem 10.6 Calculate the safe compressive load on a hollow cast iron column (both ends fixed) of 150 mm external diameter and 100 mm internal diameter and 10 m in length. Use Euler's formula with a factor of safety of 2 and $E = 95 \text{ K N/mm}^2$.

Solution: $D = 150 \text{ mm}$ $d = 100 \text{ mm}$

$$L = \frac{l}{2} = \frac{10}{2} = 2 \text{ m} = 5 \times 1000 \text{ mm}$$

$$= 5000 \text{ mm}$$

$$I = \frac{\pi}{64} (150^4 - 100^4) = 1.994 \times 10^7 \text{ mm}^4$$

$$P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 95 \times 1.994 \times 10^7}{(5000)^2} = 747.84 \text{ kN}$$

$$\text{Safe load} = \frac{747.84}{2} = 373.92 \text{ kN}$$

Problem 10.7 A hollow alloy tube 5 meter long with external and internal diameters equal to 40 mm and 25 mm respectively was found to extend by 6.4 mm under a tensile load of 60 kN. Find the buckling load for the tube, when used as a column with both ends fixed. Also find the safe compressive load for the tube, with a factor of safety of 4.

Solution: Area of the section $A = \frac{\pi}{4} (40^2 - 25^2) \text{ mm}^2 = 765.76 \text{ mm}^2$

When subjected to tension

$$\text{Extension } \delta = \frac{Wl}{AE}$$

$\therefore E = \frac{WL}{A\delta} = \frac{60 \times 10^3 \times 5 \times 10^3}{765.76 \times 6.4} = 61213.6 \text{ N/mm}^2$

When used as a column $I = \frac{\pi}{64} (40^4 - 25^4) = 106489 \text{ mm}^4$

Buckling load $P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 61213.6 \times 106489}{\left(\frac{5000}{2}\right)^2} = 10293.72 \text{ N}$

Safe load, with a factor of safety of 4

$$= \frac{10293.72}{4} = 2573.43 \text{ N}$$

Problem 10.8 A bar of length 4 meters when used as a simply supported beam and subjected to a uniformly distributed load of 60 kN/m over the whole span deflects 15 mm at the center. Determine the crippling loads when it is used as a column with the following conditions:

- (i) Both ends pinjointed,
- (ii) One end fixed and the other hinged,
- (iii) Both ends fixed.

Solution: Analysis as a beam $w = 60 \text{ kN/m} = 60 \text{ N/mm}$

$$l = 4000 \text{ mm} \quad \delta = 15 \text{ mm}$$

$$\delta = \frac{5}{384} \frac{wl^4}{EI}$$

$$\therefore EI = \frac{5}{384} \cdot \frac{wl^4}{\delta} = \frac{5}{384} \times \frac{60 \times (4000)^4}{15} = \frac{4}{3} \times 10^{13} \text{ Nmm}^2.$$

Analysis as a column

(i) When both ends are hinged

$$\begin{aligned} \text{Crippling } P &= \frac{\pi^2 EI}{l^2} = \pi^2 \times \frac{4}{3} \times 10^{13} \times \frac{1}{(4000)^2} \\ &= 8224670 \text{ N} = 8224.67 \text{ kN} \end{aligned}$$

(ii) when one end is fixed and the other end is hinged.

$$\text{Crippling load } P = \frac{2\pi^2 EI}{l^2} = 2 \times 8224.67 = 16449.34 \text{ kN}$$

(iii) when both ends are fixed

$$\text{Crippling load } P = \frac{4\pi^2 EI}{l^2} = 4 \times 8224.67 = 32898.68 \text{ kN}$$

Problem 10.9 A steel bar of rectangular section 30 mm × 40 mm one end fixed and other is free at each end is subjected to axial compression. The bar is 1.75 m long. Determine the buckling load and the corresponding axial stress using Euler's formula.

Determine the minimum length for which Euler's equation may be used to determine the buckling load, if the proportional limit of the material is 200 N/mm². Take $E = 2 \times 10^5 \text{ N/mm}^2$.

Solution: Least moment of inertia of the section of the bar

$$= I = \frac{40 \times 30^3}{12} = 90000 \text{ mm}^4$$

Since one end is fixed and other is free

Effective length = $L = 2l = 3500 \text{ mm}$

$$\begin{aligned} \text{Euler's critical load} = P &= \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2 \times 10^5 \times 90000}{(3500)^2} \text{ N} \\ &= 14502.25 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Corresponding axial stress} = \frac{P}{A} &= \frac{14502.25}{30 \times 40} \text{ N/mm}^2 \\ &= 12.09 \text{ N/mm}^2 \end{aligned}$$

We know by Euler's formula

$$\begin{aligned} P &= \frac{\pi^2 EI}{l^2} = \frac{\pi^2 EAK^2}{l^2} \\ \frac{P}{A} &= \frac{\pi^2 E K^2}{L^2} = \frac{\pi^2 E}{\left(\frac{2l}{K}\right)^2} \end{aligned}$$

Taking $\frac{P}{A} = 200 \text{ N/mm}^2$ and $E = 2 \times 10^5 \text{ N/mm}^2$

$$\left(\frac{2l}{K}\right)^2 = \frac{\pi^2 \times 2 \times 10^5}{200}$$

$$\therefore \frac{2l}{K} = 99.34 = \frac{l}{K} = \frac{99.34}{2} = 49.67$$

This means Euler's formula should not be used when the length of the member is less than 49.67 times the least radius of gyration.

Problem 10.10 Determine the ratio of the buckling strengths of two circular columns – one hollow and the other solid. Both the columns are made of the same material and have the same length, cross-sectional area and end conditions. The internal diameter of the hollow column is half of its external diameter.

Solution:

d_1 = External diameter of the hollow column

$d_2 = 0.5 d_1$ = Internal diameter of the hollow column

d = diameter of the solid column.

Since both the columns have the same area

$$\frac{\pi}{4}(d_1^2 - 0.25d_1^2) = \frac{\pi d^2}{4}$$

$$\frac{d_1^2}{d^2} = \frac{4}{3}$$

$$\text{Buckling load for the hollow column} = P_h = \frac{\pi^2 EI_h}{L^2}$$

$$\text{Buckling load for the solid column} = P_s = \frac{\pi^2 EI_s}{L^2}$$

$$\frac{P_h}{P_s} = \frac{I_h}{I_s} = \frac{d_1^4 - d_2^4}{d^4} = \frac{d_1^4 - \frac{1}{16}d_1^4}{d^4}$$

$$= \frac{15}{16} \cdot \frac{d_1^4}{d^4} = \frac{15}{16} \cdot \left(\frac{4}{3}\right)^2 = \frac{5}{3}$$

Problem 10.11 A round steel rod of diameter 30 mm and length 2 meter is subjected to a gradually increasing axial compressive load. Using Euler's formula find the buckling load. Find also the maximum lateral deflection corresponding to the buckling condition. Both ends of the rod may be taken as hinged. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$ and the yield stress of steel = 250 N/mm².

Solution: Area of the rod

$$= A = \frac{\pi d^2}{4} = \frac{\pi}{4} \times 30^2 = 706.5 \text{ mm}^2$$

Moment of the inertia of the section

$$= I = \frac{\pi d^4}{64} = \frac{\pi \times 30^4}{64} = 39740.625 \text{ mm}^4$$

Since both ends of the member are hinged, the effective length

$$= l = 2000 \text{ mm}$$

∴ Buckling load (using Euler's formula)

$$= P = \frac{\pi^2 \times 2.1 \times 10^5 \times 39740.625}{(2000)^2} \text{ N}$$

$$= 20570.9 \text{ N}$$

Direct compressive stress

$$= P_0 = \frac{P}{A}$$

$$= \frac{20570.9}{706.5} \text{ N/mm}^2$$

$$= 29.12 \text{ N/mm}^2$$

Let the maximum bending stress corresponding to buckling condition be P_b .

$$\therefore P_b + P_0 = \text{yield stress}$$

$$\therefore P_b = 250 - 29.12 = 220.88 \text{ N/mm}^2$$

Let M be the maximum bending moment which occurs at the centre

$$\therefore \frac{M}{I} = \frac{P_b}{\left(\frac{d}{2}\right)}$$

$$\therefore M = \frac{220.88}{15} \times 39740.625 = 585193.95 \text{ N mm}$$

Let the maximum central deflection be 'a' mm

$$\therefore M = Pa = 20570.9a = 585193.95$$

$$\therefore a = \frac{585193.95}{20570.9} = 28.45 \text{ mm}$$

Problem 10.12 A steel column of I-section ISHB 300 is used as a column 8 m long with both ends hinged. Determine Euler's critical load for the column. If the section is strengthened by a cover plate 310 mm × 8 mm for each flange what would be Euler's critical load?

Take for the I section area = 7485 mm²; $I_{xx} = 1.2545 \times 10^8 \text{ mm}^4$; $I_{yy} = 2.1936 \times 10^7 \text{ mm}^4$. Modulus of elasticity $E = 2 \times 10^5 \text{ N/mm}^2$.

Solution: (i) When the I-section is used as a column

Least moment of inertia of the section

$$= I = I_{yy} = 2.1936 \times 10^7 \text{ mm}^4$$

$$\text{Euler's critical load} \quad P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2 \times 10^5 \times 2.1936 \times 10^7}{(8000)^2} = 0.677 \times 10^6 \text{ N}$$

$$= 677 \text{ kN}$$

(ii) When the I section with cover plates is used as a column

$$I_{yy} = 2.1936 \times 10^7 + 2 \times \frac{8 \times 310^3}{12} = 6.1657 \times 10^7 \text{ mm}^4$$

Obviously least moment of inertia of the section

$$= I = I_{yy} = 6.1657 \times 10^7 \text{ mm}^4$$

Euler's critical load

$$P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2 \times 10^5 \times 6.1657 \times 10^7}{(8000)^2}$$

$$= 1901.65 \text{ kN}$$

Problem 10.13 Find Euler's crushing load for a hollow cylindrical cast-iron column, having 150 mm external diameter and 20 mm thickness, if it is 6 m long and hinged at both ends. $E = 80 \text{ GPa}$. Compare this load with the crushing load as given by Rankine's formula, using constants $\sigma_c = 550 \text{ MPa}$ and $\alpha = \frac{1}{1600}$. For what length of strut of this

cross-section does the Euler formula cease to apply?

Solution:

$$A = \frac{\pi}{4} [(0.15)^2 - (0.11)^2] = 81.7 \times 10^{-4} \text{ m}^2$$

$$I = \frac{\pi}{64} [D^4 - d^4] = \frac{\pi}{64} [(0.15)^4 - (0.11)^4] = 17.66 \times 10^{-6} \text{ m}^4$$

$$K^2 = \frac{I}{A} = \frac{17.66 \times 10^{-6}}{81.7 \times 10^{-4}} = 21.6 \times 10^{-4} \text{ m}^2$$

Euler's load is given by

$$P_e = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 80 \times 10^9 \times 17.66 \times 10^{-6}}{6^2} = 387327.14 \text{ N}$$

According to Rankine's formula

$$P_R = \frac{\sigma_c \cdot A}{1 + \alpha \left(\frac{l_e}{k} \right)^2} = \frac{550 \times 10^6 \times 81.7 \times 10^{-4}}{1 + \frac{1}{1600} \times \frac{6^2}{21.6 \times 10^{-4}}} = 393591.24 \text{ N}$$

The stress calculate from Euler's formula cannot exceed the yield stress of 550 MPa.

$$550 \times 10^6 = \frac{P_e}{A} = \frac{\pi^2 EI}{A L^2} = \frac{\pi^2 \times 80 \times 10^9 \times 21.6 \times 10^{-4}}{L^2}$$

$$\therefore L = 1.761 \text{ m}$$

Problem 10.14 A straight length of steel bar, 1.5 m long and 20 mm $\times$ 5 mm section, is compressed longitudinally until it buckles. Assuming Euler's formula to apply to this case, estimate the Maximum central deflection before the steel passes the yield point at 320 N/mm^2 . Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution:

$$I_{\min} = \frac{20 \times (5)^3}{12} = \frac{1 \times 10^4}{48} \text{ mm}^4 = \frac{10^{-8}}{48} \text{ m}^4$$

$$P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times 10^4}{48 \times (1.5 \times 10^3)^2} = 191.9 \text{ N}$$

There will be no deflection till the Euler load is reached. Let δ = central deflection
Maximum bending moment,

$$M = P \cdot \delta = 191.9 \cdot \delta$$

Direct stress is given by,

$$\sigma_1 = \frac{P}{A} = \frac{191.9}{20 \times 5} = 1.919 \text{ N/mm}^2$$

Bending stress is given by,

$$\sigma_2 = \frac{M}{Z} = \frac{191.9\delta \times 6}{20 \times 5^2} = \delta \times 2.3028 \text{ N/mm}^2 \left[\because z = \frac{bd^2}{6} \right]$$

Total stress

$$3.20 = 1.919 + 2302.8 \cdot \delta$$

$$\therefore \delta = 138.1 \text{ mm}$$

Problem 10.15 A built-up beam shown in fig 9.6 is simply supported at its ends. Compute its length given that when it is subjected to a load of 40 kN/m, it deflects by 10 mm. find out the safe load if this beam is used as a column with both ends fixed. Assume a factor of safety of 4. Use Euler's formula. $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution:

$$I_{xx} = \frac{1}{12} (300 \times 110^3 - 280 \times 1000^3) = 9.94166 \times 10^9 \text{ mm}^4.$$

$$I_{yy} = 2 \times \frac{50 \times (300)^3}{12} + \frac{1000 \times (20)^3}{12} = 2.25666 \times 10^8 \text{ mm}^4.$$

Let L be the length of the beam

Deflection at the centre is given by

$$\delta = \frac{5}{384} \times \frac{WL^3}{EI} = \frac{5 \times 40 \times L^4}{384 \times 2.1 \times 10^5 \times 9.94166 \times 10^8} = 10 \text{ mm}$$

$$\therefore L = 14149.6 \text{ mm} = 14.1496 \text{ m}$$

$$\text{Crippling load, } P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times 2.25666 \times 10^8 \times 4}{(14149.6)^2}$$

$$= 9344516.5 \text{ N}$$

$$\text{Safe load} = \frac{9344516.5}{4} = 2336129.1 \text{ N}$$

Problem 10.16 Determine the ratio of buckling strengths of two columns, one hollow and the other solid. Both are made of the same material and have the same length, cross-sectional area and end conditions. The internal diameter of hollow column is half of its external diameter.

Solution:

Let d_s = Diameter of solid column
 d_H = External diameter of hollow column

$$\therefore \text{Internal diameter of hollow column} = \frac{d_H}{2}$$

Area of solid column = Area of hollow column

$$\frac{\pi}{4} d_s^2 = \frac{\pi}{4} \left[d_H^2 - \left(\frac{d_H}{2} \right)^2 \right]$$

$$\text{or } d_s^2 = \frac{3}{4} \times d_H^2 \quad \therefore d_s = \frac{\sqrt{3}}{2} \cdot d_H$$

Buckling load of a column as given by Euler's formula is $P = \frac{\pi^2 EI}{L^2}$

As both the columns are having the same length, material, cross-sectional area and same ends conditions, the value of A, E and L are same for both.

$$\therefore P \propto I$$

Let P_s = Buckling load for solid column

P_H = Buckling load for hollow column

$$\therefore \frac{P_H}{I_H} = \frac{P_s}{I_s} \text{ or } \frac{P_H}{P_s} = \frac{I_H}{I_s} = \frac{\frac{\pi}{64} \left[d_H^4 - \left(\frac{d_H}{2} \right)^4 \right]}{\frac{\pi}{64} d_s^4} = \frac{15 d_H^4}{16 d_s^4}$$

$$\therefore \frac{P_H}{P_s} = \frac{15}{16} \times \left(\frac{d_H^2}{d_s^2} \right)^2 = \frac{15}{16} \times \left(\frac{4}{3} \right)^2 = \frac{15}{16} \times \frac{16}{9} = \frac{5}{3}$$

$$\therefore \frac{\text{Buckling load of hollow column}}{\text{Buckling load of solid column}} = \frac{5}{3}$$

Problem 10.17 A hollow cylindrical cast iron column is 4 m long with both ends fixed. Determine the minimum diameter of the column if it is to carry a safe load of 500 kN with a factor of safety 5. Take the internal diameter as 0.8 times the external diameter.

Take $\sigma_c = 550$ MPa and $\alpha = \frac{1}{1600}$.

Solution:

Crippling load, $P = 500 \times 10^3 \times 5 = 250 \times 10^4$ N

$$\text{Area, } A = \frac{\pi}{4} [d^2 - (0.8 d)^2] = \pi \times 0.09 d^2$$

$$\text{Moment of inertia, } I = \frac{\pi}{64} [d^4 - (0.8 d)^4] = 0.009225 \pi d^4$$

$$\therefore K = \left(\frac{I}{A} \right)^{\frac{1}{2}} = \left[\frac{0.009225 \pi d^4}{\pi \times 0.09 d^2} \right]^{\frac{1}{2}} = 0.32 d$$

Using the relation, $P = \frac{\sigma_c \cdot A}{1 + \alpha \left(\frac{l_e}{k} \right)^2}$

$$250 \times 10^4 = \frac{550 \times \pi \times 0.09 d^2}{1 + \frac{1}{1600} \left(\frac{2}{0.32d} \right)^2}$$

or $16084.4 = \frac{d^2}{1 + \frac{0.024414}{d^2}} = \frac{d^4}{d^2 + 0.0244}$

or $d^4 - 16084.4 d^2 - 392.46 = 0$
 $d^2 = \frac{16084.4 \pm \sqrt{(16084.4)^2 + 4 \times 392.46}}{2}$

$$d^2 = 16084.45$$

$$d = 126.82 \text{ mm}$$

$\therefore$ Internal diameter $= 0.8 \times 126.82 = 101.46 \text{ mm}$

Problem 10.18 A ISMB 250 Rolled Steel Joist is to be used as a column 4 meter long with both end fixed, find the safe axial load on the column allowing a factor of safety of

3. Take $\sigma_c = 320 \text{ N/mm}^2$ and $\alpha = \frac{1}{7500}$. Properties of column section are as follows:

$$\text{Area} = 4755 \text{ mm}^2$$

$$I_{xx} = 5.1316 \times 10^7 \text{ mm}^2$$

$$I_{yy} = 3.345 \times 10^6 \text{ mm}^4$$

Solution: Effective length of the column

$$L = \frac{l}{2} = \frac{4000}{2} \text{ mm}$$

$$K_2 = \frac{I_{yy}}{A} = \frac{3.345 \times 10^6}{4755} = 703.47$$

Crippling load $= P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}}$
 $= \frac{320 \times 4755}{1 + \frac{1}{7500} \times \frac{4000 \times 4000}{4 \times 703.47}} = 865456.9 \text{ N}$

Allowing a factor of a safety of 3,

Safe load on the column $= \frac{865456.9}{3} = 288485.6 \text{ N}$

Problem 10.19 A short length of tube 40 mm internal diameter and 50 mm external diameter, failed in compression at a load of 480 kN. When a 2 meters length of the same

tube was tested as a strut with fixed ends, the load at failure was 200 kN. Assuming that σ_c in Rankine's formula is given by the first test find the value of the constant α in the same formula. Find also the crippling load of this tube if it is used as a strut 3 meters long with one end fixed and the other end hinged.

Solution:

$$D = 50 \text{ mm}; d = 400 \text{ mm}$$

$$A = \frac{\pi}{4}(50^2 - 40^2) = 255 \pi = 706.86 \text{ mm}^2$$

$$I = \frac{\pi}{64}(50^4 - 40^4) = 57656.25 \pi \text{ mm}^4$$

$$K^2 = \frac{I}{A} = \frac{57656.25 \pi}{225 \pi} = 256.25 \text{ mm}^2$$

Crushing stress as per first test

$$= \sigma_c = \frac{480 \times 10^3}{225 \pi} = 679.4 \text{ N/mm}^2$$

Corresponding to the second test,

$$L = \frac{2000}{2} = 1000 \text{ mm}$$

$$P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}} = 200 \times 10^3$$

$$\frac{339.53 \times 225 \pi}{1 + \alpha \times \frac{1000^2}{256.25}} = 200 \times 10^3$$

$$1 + \alpha \times \frac{1000^2}{256.25} = 1.923 \quad \therefore \alpha = 2.365 \times 10^{-4}$$

for the column of 3 meter length

$$L = \frac{l}{\sqrt{2}} = \frac{3000}{\sqrt{2}} = 1500 \sqrt{2} \text{ mm}$$

$$P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}} = \frac{679.4 \times 225 \pi}{1 + 2.365 \times 10^{-4} \times \frac{1500^2 \times 2}{256.25}}$$

$$= 93146 \text{ N} = 93.146 \text{ kN}$$

Problem 10.20 Find Euler's critical load for a hollow cylindrical cast iron column 250 mm external diameter and 25 mm thick, if it is 6 meter long and hinged at both ends. Take $E = 8 \times 10^4 \text{ N/mm}^2$.

Compare Euler's critical load with the Rankine's critical load taking $f_c = 550$

$$\text{N/mm}^2 \text{ and } \alpha = \frac{1}{1600}$$

For what length of the column would the critical loads by Euler's and Rankine's formula be equal?

Solution: $D = 250 \text{ mm}; d = 200 \text{ mm}$

$$A = \frac{\pi}{4} (250^2 - 200^2) = 17662.5 \text{ mm}^2$$

$$I = \frac{\pi}{64} (250^4 - 200^4) = 1.67 \times 10^8 \text{ mm}^4$$

$$K^2 = \frac{I}{A} = \frac{1.67 \times 10^8}{17662.5} = 9455.06 \text{ mm}^2$$

Since both ends are hinged

Effective length $L = l = 6000 \text{ mm}$

$$\begin{aligned} \text{Euler's critical load } P_e &= \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 8 \times 10^4 \times 1.67 \times 10^8}{(6000)^2} \\ &= 3.66 \times 10^6 \text{ N} \end{aligned}$$

$$\text{Rankine's critical load } P_r = \frac{\sigma_c A}{1 + \alpha \frac{l^2}{K^2}} = \frac{550 \times 17662.5}{1 + \frac{1}{1600} \times \frac{6000^2}{9455.06}} = 2.87 \times 10^6 \text{ N}$$

Ratio of Euler's critical load to Rankine's critical load

$$= \frac{3.66 \times 10^6}{2.87 \times 10^6} = 1.275$$

For Euler's critical load to be equal to Rankine's critical load

$$\begin{aligned} \frac{\pi^2 EI}{l^2} &= \frac{\sigma_c A}{1 + \alpha \frac{l^2}{K^2}} \\ \frac{\pi^2 \times 8 \times 10^4 \times 1.67 \times 10^8}{l^2} &= \frac{550 \times 17662.5}{1 + \frac{1}{1600} \times \frac{l^2}{9455.06}} \end{aligned}$$

$$1 + \frac{l^2}{15.13 \times 10^6} = 7.37 \times 10^{-8} l^2$$

$$l = 11466.15 \text{ mm}$$

Problem 10.21 Find the greatest length of mild steel rod $25\text{mm} \times 25 \text{ mm}$ which can be used as a compression member with one end fixed and the other end free to carry a working load of 70 kN. Allow a factor of safety of 4. Take $\alpha = \frac{1}{7500}$ and $f_c = 320 \text{ N/mm}^2$.

$$\text{Solution: Area of the rod} = 25 \times 25 = 625 \text{ mm}^2$$

$$\text{Moment of inertia of the section} = I = \frac{25^4}{12} = 32552 \text{ mm}^4$$

$$K^2 = \frac{I}{A} = \frac{32552}{625} = 52.08 \text{ mm}^2$$

$$\text{Crippling load} = P = 70 \times 4 = 280 \text{ kN}$$

$$= \frac{f_c A}{1 + \alpha \frac{L^2}{K^2}}$$

$$280 \times 10^3 = \frac{320 \times 625}{1 + \frac{1}{7500} \times \frac{L^2}{52.08}}$$

$$1 + \frac{L^2}{7500 \times 52.08} = \frac{320 \times 625}{280 \times 10^3} = 1.2184$$

$$\frac{L^2}{7500 \times 52.08} = 0.214$$

$$L^2 = 0.214 \times 7500 \times 52.08 = 83588.4$$

$$\therefore L = 289.12 \text{ mm}$$

Since one end of the rod is fixed and the other end is hinged

$$\text{Effective length} \quad L = \frac{l}{\sqrt{2}}$$

$$\therefore l = \sqrt{2} L = \sqrt{2} \times 289.12 \text{ mm} = 408.81 \text{ mm}$$

Problem 10.22 A hollow cylindrical cast iron column is 4 meters long, both ends being fixed. Design the column to carry an axial load of 250kN. Use Rankine's formula and adopt a factor of safety of 5. The internal diameter may be taken as 0.80 times the external diameter. Take $\sigma_c = 550 \text{ N/mm}^2$ and $\alpha = \frac{1}{1600}$.

Solution: Let the external diameter be d mm

$$\text{Internal diameter} = d_2 = 0.8 d \text{ mm}$$

$$\text{Area of the section} = \frac{\pi}{4} (d^3 - d_2^3) = \frac{\pi}{4} (d^2 - 0.64 d^2) = 0.09 \pi d^2 \text{ mm}^2$$

$$\text{Moment of inertia of the section} = I = \frac{\pi}{64} (d^4 - d_2^4) \text{ mm}^4$$

$$\therefore K^2 = \frac{I}{A} = \frac{\frac{\pi}{64} (d^4 - d_2^4)}{\frac{\pi}{4} (d^2 - d_2^2)} \text{ mm}^2$$

$$\therefore K^2 = \frac{d^2 + d_2^2}{16} = \frac{d^2 + 0.64 d^2}{16} = \frac{1.64}{16} d^2 = 0.1025 d^2 \text{ mm}^2$$

$$\text{Safe load on the column} = 250 \text{ kN}$$

$$\therefore \text{Crippling load} = \text{Safe load} \times \text{factor of safety} \\ = 250 \times 5 = 1250 \text{ kN}$$

$$\text{Effective length of the column} = L = \frac{l}{2} = \frac{4000}{2} = 2000 \text{ mm}$$

Applying Rankine's formula

$$P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}}$$

$$1250 \times 1000 = \frac{550 \times 0.09 \pi D^2}{1 + \frac{1}{1600} \times \frac{2000 \times 2000}{0.1025 D^2}}$$

$$1250 \times 1000 = \frac{155.5 D^4}{D^2 + 24390}$$

$$155.5 D^4 = 1250000 D^2 + 1250000 \times 24390$$

$$D^4 = 8038.58 D^2 + 1.9606 \times 10^8$$

$$D^4 - 8038.58 D^2 - 1.9606 \times 10^8 = 0$$

Solving as a quadratic in D^2 ; $D^2 = \frac{8038.58 + \sqrt{8038.58^2 + 4 \times 1.9606 \times 10^8}}{2}$

$$D^2 = 18586.8$$

$$D = 136.3 \text{ mm}$$

Internal diameter

$$= 0.8 \times 136.3 = 109.04 \text{ mm}$$

Problem 10.23 The following particulars refer to an engine cylinder:

Diameter of the cylinder = 400 mm

Steam pressure in cylinder = 0.60 N/mm²

Distance between the piston and cross head = 1.25 m. Find the diameter of the piston rod allowing a factor of safety of 4. Assume that the piston is firmly fixed to the piston and

the cross head. Take $\sigma_c = 330 \text{ N/mm}^2$ and $\alpha = \frac{1}{7500}$.

Solution: Effective length of the piston rod

$$= L = \frac{1.25}{2} = 0.625 \text{ m} = 625 \text{ mm}$$

Compressive load on the piston = Steam pressure in the cylinder $\times$ area of the cylinder

$$= 0.60 \times \frac{\pi}{4} \times 400^2 = 75398 \text{ N}$$

P = Crippling load for the piston = 75398 $\times$ 4 = 301592 N

$$P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}} = 301592 \text{ N}$$

$$K^2 = \frac{I}{A} = \frac{\pi d^4}{64} + \frac{\pi d^2}{4} = \frac{D^2}{16}$$

$$\therefore \frac{330 \times \frac{\pi d^2}{4}}{1 + \frac{1}{7500} \times \frac{625^2 \times 16}{D^2}} = 301592$$

$$259.18 D^2 = 301592 \left(1 + \frac{833.33}{D^2} \right)$$

$$D^2 = 1163.64 \left(1 + \frac{833.33}{D^2} \right)$$

$$\therefore D^4 - 1163.64 D^2 - 833.33 = 0$$

$$d_2 = \frac{1163.64 + \sqrt{1163.64^2 + 4 \times 833.33}}{2} = 1164.36$$

$$\therefore d = 34.12 \text{ mm}$$

Problem 10.24 A hollow C.I. column whose outside diameter is 200 mm has a thickness of 20 mm. It is 4.5 meters long and is fixed at both ends. Calculate the safe load by Rankine's formula using a factor of safety of 4. Calculate the slenderness ratio and the ratio of Euler's and Rankine's Critical loads. For cast iron take $\sigma_c = 600 \text{ N/mm}^2$ and $\alpha = \frac{1}{1600} E = 8 \times 10^4 \text{ N/mm}^2$.

Solution: $d_1 = \text{Outer diameter} = 200 \text{ mm}$
 $d_2 = \text{Inner diameter} = 200 - 40 = 160 \text{ mm}$

Area $= A = \frac{\pi}{4} (d_1^2 - d_2^2) = \frac{\pi}{4} (200^2 - 160^2) = 11310 \text{ mm}^2$

Moment of inertia of section $I = \frac{\pi}{64} (d_1^4 - d_2^4) = \frac{\pi}{64} (200^4 - 160^4) = 4.637 \times 10^7 \text{ mm}^4$

Let $K = \text{radius of gyration}$ $K^2 = \frac{I}{A} = \frac{4.637 \times 10^7}{11310} = 4100 \text{ mm}^2$

$$\therefore K = \sqrt{4100} = 64.03 \text{ mm}$$

Effective length $= \frac{15000}{2} = 7500 \text{ mm}$

$$\therefore \text{Slenderness ratio} = \frac{7500}{64.03} = 35.14$$

Rankine's critical load $= P_r = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}}$

For cast iron $\sigma_c = 600 \text{ N/mm}^2$ and $\alpha = \frac{1}{1600}$

$$P_r = \frac{600 \times 11310}{1 + \frac{1}{1600} \times \frac{7500 \times 7500}{4100}} = 708743 \text{ N}$$

Safe load $= \frac{708743}{4} = 177185.75 \text{ N}$

Euler's critical load $P_c = \frac{\pi^2 EI}{L^2}$

Taking $E = 8 \times 10^4 \text{ N/mm}^2$

$$P_c = \frac{\pi^2 \times 8 \times 10^4 \times 4.637 \times 10^7}{7500 \times 7500} = 650885 \text{ N}$$

$$\frac{\text{Euler's Critical load}}{\text{Rankine's Critical load}} = \frac{650885}{708743} = 0.92$$

Problem 10.25 A 1.5 meter long column has a circular cross-section of 50 mm diameter. One end of the column is fixed in direction and position and the other end is free. Taking a factor of safety of 3, calculate the safe load using ,

- (i) Rankine's formula : Take $\sigma_c = 560 \text{ N/mm}^2$ and
- (ii) Euler's formula : Young's modulus for cast iron
Take $E = 2.1 \times 10^5 \text{ N/mm}^2$.

Solution : (i) By Rankine's formula

$$\begin{aligned} d &= 50 \text{ mm} \\ \text{effective length } L &= 2l = 2 \times 1.5 = 3\text{m} = 3000 \text{ mm} \\ A &= \frac{\pi}{4} \times 50^2 = 1963.5 \text{ mm}^2 \\ K^2 &= \frac{\pi d^4 / 64}{\pi d^2 / 4} = \frac{d^2}{16} = \frac{50 \times 50}{16} = 156.25 \text{ mm}^2 \\ P &= \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}} = \frac{560 \times 1963.5}{1 + \frac{1}{1600} \times \frac{3000 \times 3000}{156.25}} = 29718 \text{ N} \\ \therefore \text{ Safe load} &= \frac{29718}{3} = 9906 \text{ N} \end{aligned}$$

$$(ii) \text{ By Euler's formula } P = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 2.1 \times 10^5 \times \pi \times 50^4}{3000 \times 3000 \times 64} = 70544.9 \text{ N}$$

$$\therefore \text{ Safe load} = \frac{70544.9}{3} = 23514.97 \text{ N}$$

Problem 10.26 A hollow cast iron column 5 m long is fixed at both ends has an external diameter of 300 mm. the column supports an axial load of 1200 kN. Find the internal diameter of the column, adopting a factor of safety of 5. Take $\sigma_c = 550 \text{ N/mm}^2$ and

$$\alpha = \frac{1}{1600}.$$

$$\text{Solution : Effective length of the column } L = \frac{l}{2} = \frac{5000}{2} = 2500 \text{ mm}$$

$$\text{Area } A = \frac{\pi}{4} (300^2 - d^2), I = \frac{\pi}{64} (300^4 - d^4)$$

$$K^2 = \frac{I}{A} = \frac{300^2 + d^2}{16}$$

$$\frac{L^2}{K^2} = \frac{2500^2 \times 16}{300^2 + d^2}$$

$$\begin{aligned} P = \text{Crippling load} &= \text{Safe load} \times \text{Factor of safety} \\ &= 1200 \times 5 = 6000 \text{ kN} \end{aligned}$$

$$P = \frac{\sigma_c A}{1 + \alpha \frac{L^2}{K^2}}$$

$$6000 \times 10^3 = \frac{550 \times \frac{\pi}{4} (300^2 - d^2)}{1 + \frac{1}{1600} \times \frac{2500^2 \times 16}{300^2 + d^2}}$$

$$\frac{\pi^2}{4} \times 550 (300^2 - d^2) = 6000 \times 10^3 + \frac{6000 \times 10^3}{1600} \times \frac{2500^2 \times 16}{300^2 + d^2}$$

$$300^4 - d^4 = 13889.9 (300^2 + d^2) + 8.6812 \times 10^8$$

$$300^4 - d^4 = 1.25 \times 10^9 + 13889.9 d^2 + 8.6812 \times 10^8$$

$$300^4 - d^4 = 2.11812 \times 10^9 + 13889.9 d^2$$

$$d^4 + 13889.9 d^2 - 5.98188 \times 10^9 = 0$$

$$d^2 = \frac{-13889.9 \pm \sqrt{13889.9^2 + 4 \times 5.98188 \times 10^9}}{2}$$

$$d^2 = 70708.85$$

$$d = 265.9 \text{ mm say } 266 \text{ mm}$$

Exercise

10.1 Define slenderness ratio. Explain limitations of Euler's formula.

10.2 How will you justify that Rankine's formula is applicable for all lengths of columns ranging from short to long columns.

10.3 What is concept and significance of 'equivalent length of a column'?

10.4 Explain the assumptions made in Euler's formula.

10.5 Derive the expression for Euler's formula for a column with both the ends hinged.

10.6 Calculate Euler's buckling load of a T-section, having 100 mm flange width, overall depth 80 mm and 10 mm thick all over. The column is 3 m long and hinged at both ends. Take $E = 210 \text{ GPa}$.

(Ans. 193.25 kN)

10.7 Determine Euler's buckling load for an I-section 300 x 150 x 20 mm and 5 m long which is used as a strut with both ends fixed. Take $E = 210 \text{ GPa}$.

(Ans. 3.6 MN)

10.8 Find Euler's buckling load for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thickness. The length of the column is 2.3 m and hinged at both ends. Take $E = 205 \text{ GPa}$. Also determine the buckling load by Rankine formula using constant as 335 N/mm^2 and $1/7500$.

(Ans. 16.88 kN, 17.16 kN)

10.9 Determine the ratio of the buckling strength of a solid steel to that of a hollow column of the same material and having the same cross sectional area. The internal diameter of the hollow column is $\frac{1}{2}$ of its external diameter. Both the columns are of same length and pinned at both ends.

(Ans. 3/5)

10.10 A 2 m long column has a circular cross section of 60 mm diameter. One of the ends of the column is fixed in direction and position and other end is free. Taking factor of safety as 3 calculate the safe load using

(a) Rankine's formula using constant as 550 N/mm^2 and $1/1600$.

(b) Euler's formula and for C.I. $E = 1.3 \times 10^5 \text{ N/mm}^2$.

(Ans. (a) 11.4 kN (b) 170 kN)